

Blockchain Anomaly Transaction Detection Method Based on Graph Continual Learning

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Abstract—The rapid growth of blockchain has increased exposure to threats such as phishing, Ponzi schemes, and money laundering, making anomaly detection essential for maintaining platform security. Traditional rule-based methods and static machine learning models often fall short against evolving attack strategies, while existing graph neural network (GNN) approaches, though effective in modeling transaction graphs, require costly retraining and suffer from catastrophic forgetting when new anomalies emerge. These challenges highlight the need for adaptive detection techniques that can evolve alongside adversarial behaviors while preserving knowledge of previously observed patterns. To address this problem, this paper introduces a graph-based continual learning approach for blockchain anomaly detection. The proposed framework incorporates a topology-aware weight preservation module that captures local structural dependencies and stabilizes critical parameters during training. By explicitly modeling topological importance, the method balances knowledge retention and new task adaptation, thereby mitigating catastrophic forgetting. In addition, the framework is designed for modular compatibility with different GNN backbones, requiring only minimal adaptation for specific architectures. This flexibility ensures broad applicability in real-world blockchain systems. Extensive experiments demonstrate that our approach not only achieves high detection accuracy but also maintains stability and robustness across sequential tasks, offering a scalable and effective solution for securing blockchain ecosystems against evolving transaction anomalies.

Index Terms—Anomaly detection, blockchain, continual learning, graph neural network, security.

I. INTRODUCTION

WITH the rapid growth of blockchain applications in areas such as decentralized finance (DeFi) [1], [2], [3], supply chain management [4], [5], [6], and healthcare [7], [8], [9], the security and reliability of blockchain systems have become increasingly critical. Anomalous transactions not only cause direct financial losses but also undermine user trust and regulatory compliance. Therefore, effective detection of abnormal account behaviors is a fundamental requirement for ensuring the sustainable development of blockchain ecosystems [10], [11], [12].

However, detecting anomalies in blockchain transactions presents several unique challenges. First, the scale of transaction

data is massive, with millions of records generated daily. Second, anomalous accounts constitute only a small fraction of the data, leading to severe class imbalance. Third, labeled data is scarce, as labels often rely on expert investigation and are available only with significant delays [13]. Most importantly, malicious behaviors evolve over time, as adversaries constantly adapt their strategies to evade detection [14]. These factors make traditional static detection methods insufficient and highlight the need for adaptive, continual learning approaches.

While semi-supervised and self-supervised learning can alleviate label scarcity by leveraging unlabeled data, they typically assume a static data distribution. Blockchain networks, by contrast, evolve continuously with new addresses, contracts, and transaction patterns. To effectively capture these evolving interactions, blockchain systems are often represented as graphs that encode the flow of transactions and relationships among entities.

Graph-based representations enable the use of Graph Neural Networks (GNNs), which can model relational dependencies and higher-order structures within transaction networks. GNN-based approaches have demonstrated strong performance in fraud detection, money laundering, and phishing identification [15], [16], [17]. However, most existing GNN models are trained offline and in a task-specific manner. When new anomalous behaviors emerge, retraining from scratch is often required, which is computationally expensive and impractical for large-scale blockchain systems [18], [19].

In addition, conventional GNNs suffer from *catastrophic forgetting*, where learning new tasks can overwrite previously acquired knowledge. This issue is especially pronounced in blockchain environments, where adversaries continuously introduce new fraudulent patterns. Continual learning (CL) [20] addresses this challenge by enabling models to incrementally learn from new data while preserving knowledge of prior behaviors. Although CL methods have been studied in general neural network contexts [21], [22], their application to graph-structured blockchain data remains underexplored. Recent efforts [23], [24] offer promising directions, but a fully adapted solution for blockchain anomaly detection is still lacking.

This gap motivates our work: how can we design a graph-based continual learning model that adapts to new transaction patterns while preserving knowledge from past tasks? To address this challenge, we propose a novel continual learning framework for blockchain anomaly detection. The key idea is to integrate graph neural networks with a topology-aware mechanism that explicitly captures structural information and

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TABLE I
COMPARISON OF REPRESENTATIVE BLOCKCHAIN ANOMALY DETECTION APPROACHES

Approach	Graph Construction	Graph Deep Learning	Adaptability to New Anomalies	Catastrophic Forgetting	Suitability for Blockchain
Rule-based [25], [26]	✓	×	×(static model)	×	Basic anomaly patterns only
Supervised ML [27], [28]	×	×	×	×	Requires labeled data, poor generalization to new fraud patterns
Unsupervised ML [29], [30]	✓	×	Partial (can discover new patterns)	×	Limited accuracy, needs refinement with labels
Ensemble ML [31], [32]	Partial (model-dependent)	Partial (model-dependent)	×	×	Suffers from limitations of used models
GNN-based [15]–[17]	✓	✓	×(offline training)	×	Retraining required when new attacks appear
Incremental deep graph learning (SIEGE) [23]	✓	✓	Partial (incremental embedding)	Limited	Does not address full CL setting, retraining overhead remains
Our Method	✓	✓	✓(adapts to evolving anomalies)	✓(explicit knowledge preservation)	Addresses blockchain-specific dynamics with real-world dataset

preserves parameter importance across tasks. In doing so, the model adapts to evolving data distributions while mitigating catastrophic forgetting. The main contributions of this work are summarized as follows:

- We propose a graph-based continual learning approach for blockchain anomaly detection that can effectively and reliably identify abnormal accounts in increasingly large and complex blockchain networks. The framework is designed for flexible integration with various graph neural network architectures, allowing easy adaptation to different blockchain settings while maintaining scalability and applicability in real-world environments.
- We introduce a topology-aware weight preservation module that estimates the importance of each network parameter based on both task-related performance and topological structure. By incorporating a topology aggregation mechanism, the model retains knowledge of previous tasks while learning new ones, thereby ensuring accuracy and stability in the continual learning process.
- From the perspectives of anomaly detection and continual learning, we design comparative experiments across different neural network backbones and continual learning strategies. Experimental results demonstrate that the proposed model achieves outstanding performance in both classification accuracy and stability against catastrophic forgetting.

The remainder of this article is organized as follows. Section II introduces the related work. Section III analyzes the core challenges and gives the problem definition. Section IV introduces the proposed detection model and its modules. Section V reports comprehensive experimental results, while Section VI further analyzes the findings and practical implications. Finally, Section VII concludes the paper and outlines future research directions.

II. RELATED WORK

Research on blockchain security and anomaly detection spans multiple domains, including blockchain technology, machine learning, and graph-based analysis. To provide a comprehensive

foundation for our study, this section reviews prior work in four key areas: the fundamentals of blockchain and transaction anomalies, traditional and advanced machine learning methods for anomaly detection, the application of graph neural networks in blockchain security, and continual learning techniques for evolving threat detection. The comparisons between the representative works and our method are shown in Table I.

A. Overview of Blockchain Technology and Transaction Anomalies

Blockchain technology has rapidly expanded beyond its initial use in cryptocurrencies, finding applications in diverse domains such as supply chain management [4], [5], [6], healthcare [7], [8], [9], the Internet of Things (IoT) [33], [34], [35], Decentralized Finance (DeFi) [1], [2], [3], and various other emerging areas [36], [37], [38]. This growth has spurred extensive research on understanding blockchain structures, transaction behaviors, and systemic vulnerabilities, leveraging both datasets and analytical tools. Concurrently, the openness and pseudonymous nature of blockchain have given rise to various anomalies and security threats, highlighting the critical need for effective detection methods. In this section, we first review blockchain fundamentals and analytical resources, followed by a discussion of transaction anomalies and associated detection strategies.

1) *Blockchain Basics and Analytical Tools*: Blockchain is a decentralized and distributed ledger technology that enables secure, transparent, and tamper-resistant transaction recording without central authorities [39], [40], [41], [42], [43]. Its foundational features—immutability, consensus mechanisms [44], and cryptographic verification—have driven adoption in domains ranging from digital currencies and finance to supply chain, healthcare, and IoT. Bitcoin, introduced by Nakamoto [45], established blockchain's first major application as a cryptocurrency, while Ethereum expanded the ecosystem by enabling programmable smart contracts and decentralized applications [46], [47], [48].

With blockchain's rapid growth, research has focused on analyzing large-scale transaction data to reveal user behavior, money flows, and systemic vulnerabilities. Transaction graphs

provide a powerful abstraction, treating addresses, accounts, and contracts as nodes and their transfers as edges [49], [50], [51], [52], [53]. Quantitative graph analysis has revealed structural properties such as small-world patterns, clustering, and “rich-get-richer” dynamics, while enabling deanonymization attempts by linking pseudonymous addresses to real entities [54], [55]. Ethereum studies extend this perspective by examining heterogeneous interactions among users, tokens, contracts, and DeFi protocols, highlighting both opportunities and risks in decentralized finance ecosystems [46], [47], [56].

Reliable datasets and analysis platforms are critical to advancing blockchain research. Several benchmark datasets have been developed, including the Elliptic dataset of labeled Bitcoin transactions for anti-money laundering (AML) research [57], its extended Elliptic++ version with richer graph structures [58], and Ethereum-based datasets tailored for anomaly detection and intrusion detection systems [59]. Open-source platforms such as BlockSci offer scalable support for analyzing multiple blockchains and provide tools to study privacy, economics, and fraud [60]. Recent surveys emphasize the role of graph learning for blockchain mining and anomaly detection, integrating classical data analytics with advanced graph neural methods [61]. Visualization and software tools such as DiLeNA further aid in investigating distributed ledger networks by capturing structural properties and abnormal patterns [53], [62].

2) *Blockchain Anomalies. Threats and Detection:* Despite blockchain’s inherent security guarantees, its pseudonymity and openness create fertile ground for illicit activities [10]. Studies estimate that nearly half of Bitcoin transactions in its early years were linked to illegal activities such as drug trade, ransomware, and black-market commerce [11], [63]. Mixing services, tumblers, and privacy-focused behaviors complicate detection efforts by obfuscating fund flows [54], [63]. Moreover, Ethereum’s support for programmable contracts has facilitated new classes of fraud, including large-scale Ponzi schemes that exploit smart contracts for “trustworthy” but malicious operations [12].

The blockchain ecosystem is also exposed to protocol-level and system-level attacks. Surveys highlight vulnerabilities across consensus mechanisms, P2P networking, and smart contracts, including 51% attacks, selfish mining, Sybil, eclipse, and reentrancy exploits [64], [65], [66], [67]. Privacy leakage through weak anonymity and transaction tracing further increases risk [42], [54]. Abnormal behavior awareness frameworks for public and consortium blockchains propose detection through supervised learning, statistical analysis, and heuristic clustering [68]. Comprehensive surveys provide taxonomies of anomalies and defense strategies, underscoring the need for anomaly detection to secure blockchain infrastructures [69], [70].

Detection strategies range from rule-based auditing and statistical analysis to machine learning and graph-based methods. Approaches such as BAD (Blockchain Anomaly Detection) leverage distributed metadata to capture malicious activity [71]. Sketch-based methods offer scalable detection by summarizing blockchain history into compact structures [72], while visualization and clustering techniques help uncover suspicious transaction flows [53], [62]. Deep anomaly detection frameworks

incorporating generative, discriminative, and hybrid learning have emerged to combat sophisticated cyber-attacks in blockchain environments [67]. Collectively, these efforts reflect the pressing need to reconcile blockchain’s decentralization and privacy guarantees with robust detection systems capable of mitigating fraud, cybercrime, and systemic threats [68], [69], [70].

B. Anomaly Detection With Machine Learning

The application of Machine Learning (ML) techniques [73], [74] has become paramount in identifying anomalous and fraudulent activities within blockchain networks like Bitcoin and Ethereum. ML models can learn complex patterns from transaction data, enabling the detection of illicit behaviors that are difficult to pinpoint using traditional rule-based systems [25], [26]. These approaches are broadly categorized into supervised, unsupervised, and ensemble or advanced learning methods, each with distinct advantages and challenges [75], [76].

1) *Supervised Learning:* Supervised learning methods require a labeled dataset where transactions are pre-classified as licit or illicit. These models learn a mapping function from input features to these known labels, allowing them to predict the class of new, unseen transactions. A significant challenge in this domain is the inherent scarcity of labeled data, as illicit activities are rare and often difficult to verify [13].

Numerous studies have demonstrated the effectiveness of supervised learning. Research has shown that classical algorithms like Support Vector Machines (SVM), Logistic Regression, Random Forest (RF), and k-Nearest Neighbors (kNN) can achieve high accuracy in distinguishing illicit transactions, such as those related to money laundering [77] or Bitcoin theft [78]. For instance, Alarab et al. [77] utilized an ensemble of supervised models, achieving an accuracy of 98.13% on the Elliptic dataset. Similarly, Nerurkar et al. [79] developed an ensemble of decision trees capable of categorizing multiple types of illicit entities with high accuracy, highlighting its utility for forensic analysis. GPU-accelerated implementations of models like SVM and Random Forest have also been successfully applied for fraud detection on both Bitcoin and Ethereum networks, showcasing their scalability [80].

Beyond specific fraud types, supervised learning forms the basis for broader anomaly detection frameworks. The works in [27], [28] further explore the application of these techniques for AML tasks at cryptocurrency exchanges, emphasizing their practical relevance.

2) *Unsupervised Learning:* Unsupervised learning techniques are employed when labeled data is unavailable or scarce. These methods operate without pre-existing labels, identifying anomalies by detecting data points that deviate significantly from the majority of the data based on inherent structures or patterns, such as clusters or densities [30].

Common unsupervised approaches include clustering algorithms like k-means and anomaly detection methods like SVM [81], [82]. These techniques are valuable for exploring unknown fraud patterns or for initial data analysis. Vlahavas et al. [29] applied k-means clustering directly to Bitcoin transactions (rather than addresses) using novel features,

successfully identifying distinct clusters representing common users, mining activities, exchange transactions, and a small cluster of potentially illicit activity. Their work demonstrated the value of unsupervised learning for characterizing transaction types without prior labeling.

However, a challenge noted in the literature is that purely unsupervised anomaly detection methods can sometimes be inadequate for accurately pinpointing sophisticated illicit patterns, often requiring subsequent refinement or labeling efforts [13], [30], [83]. Surveys by Shafiq et al. [84] and Cholevas et al. [30] provide a comprehensive overview of how various unsupervised algorithms and their integrations are applied to detect anomalies across different blockchain networks.

3) *Ensemble and Advanced Learning Methods*: Ensemble and advanced learning methods have shown remarkable performance in handling the complexities of blockchain anomaly detection. Hasan et al. [85] integrated explainable AI techniques with ensemble classifiers, introducing the XGBCLUS under-sampling algorithm to address data imbalance and achieving enhanced true positive rates. Nayyer et al. [31] proposed a stacking ensemble model combining Decision Tree, Naive Bayes, K-Nearest Neighbors, and Random Forest, achieving 97% accuracy and 99% AUC-ROC score. Umer et al. [32] developed an ensemble deep learning approach combining Convolutional Neural Networks and Long Short-Term Memory, with bagged LSTM achieving 96.4% accuracy. Pahuja and Kamal [86] implemented an ensemble learning framework using the CRISP-DM methodology, with LightGBM achieving 99.2% accuracy in Ethereum fraud detection. Ashfaq et al. [87] combined blockchain technology with machine learning, utilizing XGBoost and Random Forest for fraud detection while ensuring system security through smart contracts.

Advanced techniques have also been explored for specific applications: Lorenz et al. [13] addressed label scarcity through active learning, matching fully supervised performance with only 5% of labels. Huang et al. [88] developed a LightGBM model with memory components for gambling detection on Ethereum. Yuan et al. [89] employed node2vec network embedding with one-class SVM for phishing detection, achieving an F-score of 0.846. Chen et al. [90] used data mining and machine learning to detect Ponzi schemes on Ethereum. Ehsan et al. [91] enhanced Ethereum anomaly detection through sophisticated feature selection techniques combined with multiple classifiers, achieving 98% accuracy.

Recent reviews by Hisham et al. [75] and Chithanuru et al. [76] provide comprehensive examinations of machine learning and artificial intelligence techniques for blockchain anomaly detection, while Azad et al. [92] discuss future opportunities in machine learning for blockchain data analysis. The integration of these advanced methods continues to evolve, addressing challenges such as data imbalance, explainability, and real-time detection requirements in blockchain security.

C. Graph Neural Networks in Blockchain Security

The inherent graph topology of blockchain systems—where transaction networks naturally form complex relational structures—has positioned Graph Neural Networks

(GNNs) [93], [94] as transformative tools for security applications. Unlike traditional machine learning approaches that process transactions as isolated events, GNNs explicitly model the intricate topological dependencies between addresses, transactions, and smart contracts. This capability enables the detection of sophisticated attack patterns that manifest through structural anomalies in transactional networks, fundamentally advancing beyond feature-based detection paradigms.

1) *Foundational Graph Analysis for Blockchain Security*: Fundamental research has established graph-theoretic analysis as indispensable for understanding blockchain ecosystems. Chen et al. [95] pioneered this domain through comprehensive graph analysis of Ethereum, constructing three distinct graphs that revealed systemic behavioral patterns. Survey studies by Wu et al. [93] and Khan et al. [96] systematically unified methodologies for cryptocurrency transaction graph mining, while Song et al. [97] provided analysis of why blockchain systems intrinsically require graph-based approaches.

The temporal dimension of blockchain graphs received formal treatment through Lin et al.'s [18] T-EDGE framework, which introduced temporal weighted multidigraph embedding specifically for Ethereum transaction networks. Complementary work by Gaihre et al. [98] examined user anonymity behaviors through transaction graph analysis, uncovering the counterintuitive finding that wealthy addresses exhibit significantly stronger anonymity concerns than ordinary users. Liu et al. [99] extended graph analysis to blockchain documentation through heterogeneous GNNs for whitepaper analysis, demonstrating how textual and structural information can be jointly modeled.

2) *Illicit Activity Detection With GNNs*: GNN-based methods have become indispensable for combating a wide spectrum of blockchain-related crimes, ranging from phishing scams and financial fraud to money laundering and other illicit activities. These threats often share structural signatures in transaction graphs, such as coordinated account behavior, temporal bursts of suspicious transfers, or subgraph patterns designed to obscure fund flows. By modeling these relational dependencies, GNNs can identify anomalous activity that traditional feature-based or rule-driven approaches fail to capture.

For phishing and fraud detection, Chen et al. [17] established the paradigm of framing phishing detection as node classification with Graph Convolutional Networks (GCNs). Building on this, Yu et al. [100] introduced MP-GCN with enhanced message passing, Li et al. [101] proposed TTAGN for temporal transaction aggregation, and Zhang et al. [21] developed GrabPhisher with temporally evolving GNNs that adapt to new attack patterns. Higher-order dependencies have been effectively captured by Wang et al. [37] through Transaction Subgraph Networks (TSGN), while Xia et al. [102] demonstrated that attributed ego-graph embeddings improve robustness in class-imbalanced detection tasks. Targeting Bitcoin specifically, Gupta et al. [103] designed adaptive GCN frameworks achieving unprecedented 98.75% accuracy. Cross-chain detection challenges were further addressed by Yan et al. [104] with adversarial domain adaptation techniques that transfer knowledge between heterogeneous platforms.

Complementary approaches have focused on combining user-level behavior with global financial dynamics. Ghosh et al. [105]

proposed CATALOG to jointly model local and macro-level signals, while Wan et al. [106] advanced early-stage detection by leveraging feature engineering and network embeddings [107]. Tan et al. [16], [19] implemented full-stack Ethereum fraud detection systems achieving consistent 95–96% accuracy, with Lin et al. [108] further enhancing detection via Phish2vec’s temporal-heterogeneous embeddings. Together, these works illustrate how phishing and fraud campaigns can be effectively disrupted by capturing both micro-level structural anomalies and macro-level transaction flows.

For AML and illicit activity detection, Lo et al. [15] pioneered self-supervised methods with Inspection-L, integrating graph isomorphism networks with random forests for Bitcoin money laundering. Pocher et al. [109] applied Graph Attention Networks (GATs) to AML tasks, demonstrating their effectiveness in uncovering subtle relational patterns among entities. Ouyang et al. [110] developed Bit-CHetG with subgraph contrastive learning for group detection, while Nerurkar et al. [111] achieved 92% accuracy in multi-class classification of illicit Bitcoin activities using spectral GCNs. Advanced tracing frameworks such as Cheng et al.’s [112] Evolve Path Tracer enabled early-stage malice detection through dynamic asset flow analysis, and Hanan Al-Harbi [113] deployed spatial-temporal GNNs for continuous transaction monitoring. Real-time detection has also been pursued: Ibrahim et al. [114] combined GNNs with anomaly detection for streaming data environments, marking a crucial step toward operational deployment in fast-moving financial systems.

In unifying phishing, fraud, and AML detection, these approaches demonstrate how GNNs provide a holistic defense framework by capturing both opportunistic scams and large-scale organized crime. Importantly, they move beyond transaction-level analysis to reveal higher-order coordination, temporal structuring, and laundering strategies that exploit blockchain’s pseudonymous design. This convergence underscores GNNs’ potential as the backbone of next-generation blockchain security infrastructures.

3) *Advanced Applications and Emerging Paradigms:* Novel GNN applications continue to expand the frontiers of blockchain security beyond financial fraud detection. Smart contract analysis has been revolutionized through several approaches: Liu et al. [115] created heterogeneous graph transformers for abnormal contract detection; Liu et al. [116] developed evolved graph attention for dynamic attribute graphs; and Zhu et al. [117] introduced EtherShield for time-interval based malicious behavior identification.

Data provenance received significant enhancements through Geng et al.’s [118] graph attention mechanism that improves traceability, while Guo et al. [119] revolutionized covert transaction detection using graph generative networks. IoT-blockchain security was substantially strengthened by Wang et al. [120] through specialized anomaly detection frameworks that protect medical IoT systems.

Emerging research directions include Liu et al.’s [116] work on public blockchain anomaly detection with evolved graph attention, demonstrating how GNNs can adapt to continuously evolving transaction patterns. These diverse applications

underscore GNNs’ versatility across blockchain security domains—from protecting financial transactions to securing smart contracts and IoT ecosystems. As blockchain technology continues to evolve, GNN-based security frameworks are poised to address increasingly sophisticated threats through continuous learning of complex network dynamics.

D. Continual Learning Methods for Anomaly Detection

Continual learning (CL), also referred to as lifelong learning, has emerged as a critical paradigm for enabling machine learning models to adapt to new information over time without catastrophically forgetting previously acquired knowledge [121]. This capability is paramount in dynamic environments like blockchain, where new transaction patterns and attack vectors continuously emerge. This subsection reviews foundational CL techniques in neural networks, their extension to graph-structured data, and their specific applications within blockchain and anomaly detection contexts.

1) *Continual Learning in Neural Networks:* Early works in continual learning were motivated by the need to prevent catastrophic forgetting in neural networks. Kirkpatrick et al. [122] proposed elastic weight consolidation (EWC), which preserves parameters critical to prior tasks by selectively slowing their updates. Li and Hoiem [123] introduced Learning without Forgetting (LwF), enabling networks to adapt to new tasks while retaining old capabilities using only new task data. Similarly, Lopez-Paz and Ranzato [124] developed Gradient Episodic Memory (GEM), which stores past examples and replays them to balance knowledge transfer and retention. Aljundi et al. [125] proposed Memory Aware Synapses (MAS), which dynamically measures parameter importance in an unsupervised manner. Other approaches include hard attention masks that preserve task-specific knowledge [126], and meta-learning-based task-agnostic continual learning strategies [127]. Xu and Zhu [128] further extended this line with Reinforced Continual Learning, where reinforcement learning identifies optimal architectures for new tasks while preserving past performance. Surveys provide holistic overviews: Hadsell et al. [20] emphasized biologically inspired mechanisms such as modularity and memory, De Lange et al. [129] analyzed state-of-the-art methods with extensive comparisons, and Wang et al. [22] summarized objectives, taxonomies, and practical applications of CL. Foundational perspectives are also offered by Liu’s lifelong learning paradigm [121] and van de Ven et al.’s definitions of scenarios (task-, domain-, and class-incremental) [130], [131]. Collectively, these works establish the theoretical and algorithmic basis for continual learning in deep neural networks.

2) *Continual Graph Learning:* Graph-structured data presents unique challenges for continual learning due to its non-Euclidean structure and evolving topologies. Zhou and Cao [132] introduced ER-GNN, an experience replay-based framework that reuses selected node embeddings to mitigate forgetting in GNNs. Liu et al. [133] proposed Topology-aware Weight Preserving (TWP), a plug-and-play module that stabilizes graph aggregation parameters to retain structural knowledge. Sun et al. [134] addressed curvature changes

in evolving graphs with RieGrace, a self-supervised learner in adaptive Riemannian spaces that incorporates distillation without requiring labels. Su et al. [135] focused on inductive Node-wise Graph Incremental Learning, introducing Structural-Shift-Risk-Mitigation to address distribution shifts in evolving graphs. Two recent surveys provide comprehensive taxonomies and analyses: Zhang et al. [21] reviewed task settings, methods, and challenges in continual graph learning, while Tian et al. [136] categorized methods from the perspective of catastrophic forgetting and highlighted the need for continuous performance improvement. Benchmarking efforts, such as Ko et al.'s BeGIn framework, further standardize scenarios and evaluation for graph continual learning [137]. Together, these works advance the understanding of how continual learning principles can be extended to complex graph domains.

3) *Incremental Learning for Blockchain and Anomaly Detection*: Anomaly detection requires continual adaptation, as malicious behaviors evolve rapidly. Faber et al. [14] highlighted the unique challenges of lifelong anomaly detection, including shifting normal behaviors and scarce anomalies, and stressed the importance of lifelong learning to enhance robustness. In blockchain specifically, Qi [138] demonstrated the synergy between GNNs and incremental learning for fraud detection, enabling adaptation without full retraining. Li et al. [23] introduced SIEGE, a self-supervised incremental deep graph learning model for Ethereum phishing detection, leveraging pretext tasks for effective embeddings on massive unlabeled data. Neto et al. [139] addressed covert cryptocurrency mining with MineCap, a super-incremental learning framework applied to software-defined networking. Bielak et al. [24] developed FILDNE, a framework for incremental embeddings on dynamic networks, which reduces computational costs while maintaining accuracy on evolving graphs. These approaches showcase how incremental learning paradigms are increasingly tailored to security-sensitive and blockchain-specific anomaly detection tasks.

4) *Discussion and Summary*: This section reveals a progression from general CL techniques to their specialization for graph data and, more recently, for dynamic real-world applications like blockchain security and anomaly detection. While replay and regularization methods provide a strong foundation, their direct application to graphs requires careful consideration of topological dependencies and structural shifts [135]. Furthermore, the anomaly detection context adds layers of complexity due to data imbalance, evolving normality, and the critical need for stability-plasticity balance [14].

Existing continual graph learning methods like TWP [133] and ER-GNN [132] offer valuable design principles for preserving knowledge. However, their application in the blockchain domain remains limited. Most blockchain-oriented works, such as Li et al. [23], focus on incremental learning but often do not explicitly address the rigorous continual learning setting with distinct, sequential tasks and the formal mitigation of catastrophic forgetting. A significant gap exists in developing dedicated continual learning frameworks that are deeply integrated with graph-based anomaly detection for blockchain, capable of learning continuously from a sequence of emerging

TABLE II
LIST OF ACRONYMS

Acronym	Definition
AML	Anti-Money Laundering
AP	Average Performance
AF	Average Forgetting
CL	Continual Learning
CI	Class Imbalance
DeFi	Decentralized Finance
EWC	Elastic Weight Consolidation
GAT	Graph Attention Network
GCN	Graph Convolutional Network
GEM	Gradient Episodic Memory
GNN	Graph Neural Network
GIN	Graph Isomorphism Network
kNN	k-Nearest Neighbors
LwF	Learning without Forgetting
MAS	Memory Aware Synapses
ML	Machine Learning
MSAL	Masked Self-Attention Layer
RF	Random Forest
SVM	Support Vector Machine
TWP	Topology-aware Weight Preserving

fraudulent patterns without retraining from scratch. Our work, detailed in Section IV, aims to bridge this gap by proposing a novel continual learning framework built upon topology-aware weight preservation, specifically designed for the evolving graph of blockchain transactions. To enhance readability and provide a consolidated reference for the acronyms introduced throughout the paper, Table II provides a comprehensive list of key terms and their definitions.

III. PROBLEM ANALYSIS AND DEFINITION

A. Problem Analysis

Designing blockchain anomaly detection models with deep graph learning is particularly challenging due to the unique characteristics of blockchain data. Millions of transactions are generated daily, yet only a tiny fraction involve illicit activity, creating severe class imbalance. At the same time, reliable labels are scarce and often delayed, since they typically rely on investigator verification. Furthermore, blockchain environments are highly dynamic: both normal and malicious behaviors evolve over time, making static detection methods quickly outdated. A practical detection model must therefore be able to adapt continually, capturing new abnormal patterns without discarding knowledge of previously observed ones.

The major challenges can be summarized as follows:

- *Data imbalance*: In real-world blockchains, benign accounts vastly outnumber illicit ones. This imbalance biases models toward the majority class, often leading to poor recall on anomalies, which are the most critical to detect.
- *Label scarcity*: High-quality labels for anomalous behaviors are difficult to obtain, as they are based on external reports or expert investigations. Such labels cover only a small portion of accounts and are usually available with delays, creating both data sparsity and temporal imbalance.
- *Concept drift*: Normal account behaviors are not stationary; they shift over time due to changes in user habits,

seasonal patterns, or external events. Detection models that fail to adapt to these shifts risk misclassifying benign behaviors as anomalies or overlooking new types of fraud.

- *Emergence of new attacks*: Adversaries continually adjust strategies, exploiting code vulnerabilities or novel features of smart contracts. Some attacks extend known patterns, while others are entirely new. Without incremental adaptation, existing models either require full retraining—which is computationally expensive—or cannot recognize these new behaviors at all.
- *Retraining overhead*: Traditional retraining strategies demand repeated access to the full historical dataset, which is both inefficient and impractical at blockchain scale. Ideally, anomaly detection should learn from limited new samples while preserving knowledge of past attacks without revisiting all prior data.

In summary, blockchain anomaly detection requires models that are robust to imbalance and label scarcity, adaptive to evolving behaviors, and efficient enough to learn continually from new threats without incurring prohibitive retraining costs. These challenges motivate the need for continual learning frameworks tailored to graph-structured transaction data.

B. Problem Definition

The ability to acquire and refine knowledge from a continuous stream of non-stationary and imbalanced data distributions is the hallmark of the *continual learning* (also referred to as lifelong learning) paradigm. The training process of continual learning systems differs from conventional approaches: tasks are learned incrementally, and each sample can be represented by a triplet (x, y, t) , where x is the feature vector, y is the target label, and t is the task descriptor that identifies the task represented by the tuple $(x, y) \sim P_t$. In this setting, the observed data are not drawn from a fixed probability distribution $P(X, Y, T)$; instead, the data associated with sequential tasks are non-IID. Thus, the grand objective of continual learning is to construct a model $f : X \times T \rightarrow Y$ that predicts the target label y for a test pair (x, t) such that $(x, y) \sim P_t$.

In real-world blockchain environments, the number of normal accounts participating in transactions is typically much larger than that of malicious accounts. This large quantitative disparity is one of the main causes of *Class Imbalance* (CI) observed in many blockchain account datasets. Moreover, due to privacy concerns, the availability of up-to-date multi-label datasets of malicious traffic is extremely limited. For these reasons, it is challenging to formulate blockchain intrusion detection as a multi-class classification problem.

To approximate realistic traffic scenarios, we combine both normal and malicious account data to form a task t . During training on a specific task dataset $\mathcal{D}_t$, direct access to data from other tasks $\mathcal{D}_o$ ($o \neq t$) is prohibited. This leads to the following objective function:

$$\mathcal{L}(\mathcal{D}_t; \Theta) = \frac{1}{|\mathcal{D}_t|} \sum_{(x, y) \in \mathcal{D}_t} \ell(f_\Theta(x), y) \quad (1)$$

where x and y denote the feature vector and corresponding target label (normal or abnormal), f_Θ is the prediction function parameterized by Θ , and $\ell(\cdot)$ is the sample-wise loss function. However, catastrophic forgetting can degrade the quality of learning, motivating the development of various mitigation strategies. Regularization-based methods impose constraints on model parameters to prevent drastic changes during new task learning. Dynamic architecture approaches adapt the model's structure, while memory replay methods maintain a cache of representative samples from prior tasks to alleviate forgetting. Combining these strategies, the training objective in (1) can be reformulated as:

$$\mathcal{L}(\mathcal{D}_t; \Theta, \Phi) = \frac{1}{|\mathcal{D}_t|} \sum_{(x, y) \in \mathcal{D}_t} [\ell(f_\Theta(x), y) + R(\Phi)] \quad (2)$$

where $R(\Phi)$ is the regularization term introduced by the chosen catastrophic forgetting mitigation technique, and Φ collects the corresponding auxiliary variables or hyperparameters.

In the learning sequence of a blockchain anomaly detection model, the system receives a series of disjoint tasks $T = \{T_1, T_2, \dots, T_K\}$, which are learned sequentially. Each task T_k consists of a training node set V_k^{tr} , a test node set V_k^{te} , and their corresponding feature sets X_k^{tr} and X_k^{te} . Each node $v_i \in \{V_k^{\text{tr}} \cup V_k^{\text{te}}\}$ is associated with a class label $y^i \in Y_k$, where $Y_k = \{y^1, y^2, \dots, y^{c_k}\}$ is the label set and c_k is the number of classes in task T_k . In a continual learning setting, different tasks correspond to disjoint partitions of the dataset, and their label sets are non-overlapping. Once the learning of a task is completed, its data are no longer available. The objective of this paper is to learn a shared GNN model f_Θ , parameterized by $\Theta = \{\theta_r\}$, across a sequence of graph-related tasks T , such that the model performs well on new tasks while retaining knowledge of previously learned tasks.

IV. THE PROPOSED SCHEME

This section introduces the proposed graph-based continual learning framework for blockchain anomaly transaction detection. Instead of treating blockchain accounts or transactions as independent samples, we explicitly model the underlying transaction network as a graph, where anomalous behaviors such as phishing, Ponzi schemes, and other fraud patterns emerge as distinctive structural patterns. On top of this transaction graph, a Graph Neural Network (GNN) backbone is used to learn discriminative node representations for classifying normal and abnormal accounts.

To support long-term deployment in dynamic blockchain environments, where new types of anomalous behaviors and previously unseen transaction patterns continually appear, the framework integrates a Topology-aware Weight Preserving (TWP) module. TWP jointly considers (i) task-related prediction performance and (ii) blockchain-specific topological structures when estimating parameter importance. A sparse regularization mechanism is further introduced to maintain model plasticity, preventing the detector from becoming over-constrained and unable to absorb new anomalous patterns. Finally, we show how the proposed scheme can be plugged into arbitrary GNN architectures

(e.g., GCN, GIN) to support a wide range of blockchain anomaly detection scenarios. The complete pseudocode of the continual anomaly detection framework is summarized in Algorithm 1.

A. Graph Representation Learning

In the context of blockchain anomaly transaction detection, the underlying data are naturally organized as a transaction graph. We model the blockchain as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node $v \in \mathcal{V}$ corresponds to an account (e.g., user address or contract), and each edge $(u, v) \in \mathcal{E}$ represents a transfer of assets or invocation of a smart contract between accounts. Node features encode behavioral statistics, such as transaction frequency, in/out degree, token types, and temporal patterns, enabling the model to distinguish normal and anomalous accounts.

Formally, given $\mathcal{G}$ with $|\mathcal{V}| = N$ nodes, a GNN is defined by a set of node input features $\mathbf{X}^{[0]} = \{\mathbf{x}_1^{[0]}, \mathbf{x}_2^{[0]}, \dots, \mathbf{x}_N^{[0]}\}$ and an adjacency matrix $\mathbf{A}$ that represents the observed transaction relations between accounts. The hidden representation of node u at the ℓ -th layer is denoted by $\mathbf{x}_u^{[\ell]}$, where $\mathbf{x}_u^{[\ell]} \in \mathbb{R}^{F_\ell}$ and F_ℓ is the feature dimension at layer ℓ .

Graph Convolutional Networks (GCNs) compute node representations using a first-order approximation of the graph spectrum, where the adjacency matrix $\mathbf{A}$ is fixed and encodes static transaction connectivity. Graph Attention Networks (GATs) extend GCNs by introducing an attention mechanism that optimizes node representations through a weighted aggregation of neighbor features. In the blockchain scenario, this means that an account can place higher attention on neighbors with more suspicious transaction patterns (e.g., hubs connected to many scams) and lower attention on benign or weakly related neighbors. Unlike the fixed normalization operation in GCNs, GATs enable each node to selectively attend to its neighbors based on contextual information in the local transaction neighborhood, thereby reducing reliance on global structural information. By stacking Masked Self-Attention Layers (MSALs), GATs capture the characteristics of neighboring accounts and assign weights according to their importance for anomaly detection. The updated node features are then computed based on these weights.

Specifically, for node u and one of its neighbors $v \in \mathcal{N}(u)$ at layer ℓ , the attention score is first computed as

$$\alpha_{uv}^{[\ell]} = \text{Softmax}_{v \in \mathcal{N}(u)} \mathbf{a}(\mathbf{x}_u^{[\ell-1]} \mathbf{U}^{[\ell]}, \mathbf{x}_v^{[\ell-1]} \mathbf{U}^{[\ell]}) \quad (3)$$

where $\alpha_{uv}^{[\ell]}$ denotes the attention score from node u to node v , $\mathbf{a}(\cdot)$ is a learnable neural network, and $\mathbf{U}^{[\ell]}$ is the trainable weight matrix at layer ℓ .

These attention scores reflect the strength and importance of relationships between accounts in the blockchain transaction network. The normalized attention coefficient is given by

$$\tilde{\alpha}_{uv}^{[\ell]} = \frac{\exp(\text{LReLU}(\alpha_{uv}^{[\ell]}))}{\sum_{q \in \mathcal{N}(u)} \exp(\text{LReLU}(\alpha_{uq}^{[\ell]}))} \quad (4)$$

where $\tilde{\alpha}_{uv}^{[\ell]}$ represents the normalized attention score from node u to node v , and $\text{LReLU}(\cdot)$ is the LeakyReLU activation function.

Algorithm 1: Graph-Based Continual Learning for Blockchain Anomaly Account Detection.

Require: Blockchain anomaly transaction dataset $\mathcal{D}_k^{\text{tr}}$ constructed from historical blockchain records, where nodes correspond to accounts and edges encode transaction flows

- 1: Initialize model parameters Θ_1
- 2: **for** each training epoch **do**
- 3: Train the model on the first anomaly detection task using $\mathcal{D}_1^{\text{tr}}$ and compute the graph representation of the blockchain transaction graph $\mathcal{G}$
- 4: Infer node labels (normal vs. different abnormal account behaviors)
- 5: Compute cross-entropy loss for anomaly classification
- 6: Perform backpropagation and update parameters
- 7: **end for**
- 8: Obtain the optimal parameters for the first task Θ_1^*
- 9: **for** each subsequent anomaly detection task $k \in \{1, \dots, K-1\}$ **do**
- 10: Set model parameters $\Theta_{k+1} \leftarrow \Theta_k^*$ to carry over knowledge of previously seen anomalous behaviors
- 11: **for** each training epoch **do**
- 12: Train the model on task k using $\mathcal{D}_k^{\text{tr}}$, and compute the graph representation of $\mathcal{G}$ that encodes the evolving blockchain transaction network
- 13: Infer node labels of $\mathcal{G}$ to distinguish anomalous from benign accounts for the current fraud category
- 14: Compute cross-entropy loss for task T_k
- 15: Perform backpropagation
- 16: # Loss preservation (task-related anomaly detection performance)
- 17: Compute gradients on task k : $g_r(\mathbf{X}_k^{\text{tr}}) = \frac{\partial \mathcal{L}}{\partial \theta_r}$
- 18: Obtain loss-based importance scores: $\Omega_k^{(\text{err})} = [\|g_r(\mathbf{X}_k^{\text{tr}})\|]$
- 19: # Topology preservation (blockchain transaction structure)
- 20: Compute topology-based importance scores: $\Omega_k^{(\text{top})} = [\|g_r(\mathbf{Z}_k^{[\ell-1]})\|]$
- 21: # Continual learning update for evolving transaction anomalies
- 22: Compute final importance scores for task T_k : $\Omega_k = \lambda_l \Omega_k^{(\text{err})} + \lambda_t \Omega_k^{(\text{top})}$
- 23: Compute total loss: $\tilde{\mathcal{L}}_{k+1}(\Theta) = \mathcal{L}_{k+1}^{\text{tot}}(\Theta) + \beta \|\Omega_{k+1}\|_1$
- 24: **end for**
- 25: Obtain the optimal parameters for the current anomaly type Θ_{k+1}^*
- 26: **end for**

Ensure Final model parameters Θ_K^* for multi-type blockchain anomaly account detection

Finally, the updated feature representation of node u at the ℓ -th layer is computed as

$$\mathbf{x}_u^{[\ell]} = \phi \left(\sum_{v \in \mathcal{N}(u)} \tilde{\alpha}_{uv}^{[\ell]} \mathbf{x}_v^{[\ell-1]} \mathbf{U}^{[\ell]} \right) \quad (5)$$

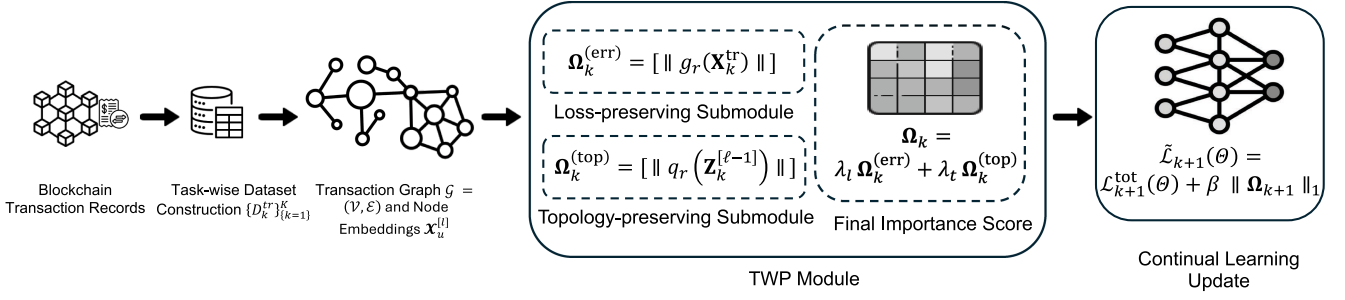


Fig. 1. Overall framework of the TWP module in blockchain anomaly account detection.

where $\phi(\cdot)$ is a nonlinear activation function. The adjacency matrix $\mathbf{A}$ encodes the neighbor set $\mathcal{N}(u)$ for each node u and thus determines the aggregation pattern, which is a core component of GNN-based anomaly detection on transaction graphs.

B. Topology-Aware Weight Preserving Module

In a blockchain setting, anomalous behaviors are often reflected not only in individual account features but also in how accounts are wired together through transaction flows (e.g., fan-in/fan-out structures, layered transfers, star-like scam clusters). The TWP module is designed to capture such topological information and identify key parameters that are crucial for both task-related anomaly classification and the preservation of meaningful transaction structures. The module consists of two main submodules: *loss-preserving*, which focuses on anomaly detection performance, and *topology-preserving*, which focuses on the structural patterns of the transaction graph. The overall framework is illustrated in Fig. 1.

1) *Loss-preserving submodule*: After training on task T_k (e.g., detecting one specific category of anomalous accounts such as phishing or Ponzi schemes), the model obtains a set of optimized parameters Θ_k^* that minimize the classification loss for that task. However, not all parameters contribute equally to correctly identifying abnormal accounts and separating them from benign ones. Therefore, it is important to identify those parameters that are particularly critical for minimizing detection loss on the blockchain dataset.

Given the training set $\mathcal{D}_k^{tr}$ of task T_k , the loss can be written as $\mathcal{L}(\mathbf{X}_k^{tr}; \Theta)$, where $\mathbf{X}_k^{tr}$ contains the node features in $\mathcal{D}_k^{tr}$, and $\Theta = \{\theta_r\}$ denotes all network parameters. The loss change under an infinitesimal perturbation $\Delta\Theta = \{\Delta\theta_r\}$ can be approximated using a first-order Taylor expansion:

$$\mathcal{L}(\mathbf{X}_k^{tr}; \Theta + \Delta\Theta) - \mathcal{L}(\mathbf{X}_k^{tr}; \Theta) \approx \sum_r g_r(\mathbf{X}_k^{tr}) \Delta\theta_r \quad (6)$$

where $\Delta\theta_r$ is the infinitesimal change of parameter θ_r , and $g_r(\mathbf{X}_k^{tr}) = \frac{\partial \mathcal{L}}{\partial \theta_r}$ is the gradient of the loss with respect to θ_r , which approximates the contribution of θ_r to the overall anomaly detection loss.

To preserve knowledge of task T_k when learning future tasks (i.e., new anomaly types or updated transaction patterns), it is desirable to stabilize parameters that are highly important for

minimizing the loss. Similar to Elastic Weight Consolidation (EWC), the importance of parameter θ_r can be estimated using the magnitude of its gradient g_r . Specifically, the loss-based importance scores of parameters for task T_k are defined as

$$\Omega_k^{(err)} = [\|g_r(\mathbf{X}_k^{tr})\|]$$

where $\Omega_k^{(err)}$ reflects the degree to which each parameter contributes to minimizing the anomaly classification loss of task T_k .

2) *Topology-preserving submodule*: This submodule aims to identify parameters that are essential for preserving learned topological information in the blockchain transaction graph. GAT is adopted as the backbone, where the attention coefficients between a central node and its neighbors are modeled as representations of local transaction topology (e.g., how suspicious funds propagate between accounts). Let $\mathbf{Z}_{u,v}^{[\ell-1]}$ denote the pairwise embeddings of nodes u and v at layer $\ell - 1$. The topology-related score between node u and its neighbor v at layer ℓ can be expressed as

$$\gamma_{uv}^{[\ell]} = \mathbf{b}(\mathbf{Z}_{u,v}^{[\ell-1]}; \Theta^{[\ell]}) \quad (7)$$

where $\mathbf{b}(\cdot)$ is a neural network, and $\Theta^{[\ell]}$ denotes the subset of parameters at layer ℓ .

Similar to the loss-preserving case, the change under an infinitesimal perturbation $\Delta\Theta^{[\ell]}$ can be approximated by

$$\begin{aligned} \mathbf{b}(\mathbf{Z}_{u,v}^{[\ell-1]}; \Theta^{[\ell]} + \Delta\Theta^{[\ell]}) - \mathbf{b}(\mathbf{Z}_{u,v}^{[\ell-1]}; \Theta^{[\ell]}) \\ \approx \sum_r q_r(\mathbf{Z}_{u,v}^{[\ell-1]}) \Delta\theta_r \end{aligned} \quad (8)$$

where $q_r(\mathbf{Z}_{u,v}^{[\ell-1]}) = \frac{\partial \mathbf{b}}{\partial \theta_r}$ is the gradient of the topology score $\gamma_{uv}^{[\ell]}$ with respect to parameter θ_r .

Unlike the scalar task loss, the topology scores around a central node form a multidimensional vector at layer ℓ . The topological loss of $\mathcal{D}_k^{tr}$ is defined as the squared ℓ_2 norm of this vector, and its gradient with respect to θ_r is computed as

$$q_r(\mathbf{Z}_{u,v}^{[\ell-1]}) = \frac{\partial}{\partial \theta_r} \left\| [\gamma_1^{[\ell]}, \gamma_2^{[\ell]}, \dots, \gamma_{|\mathcal{D}_k^{tr}|}^{[\ell]}] \right\|_2^2 \quad (9)$$

where $\mathbf{Z}_{u,v}^{[\ell-1]}$ is the output of the $(\ell - 1)$ -th layer. The importance scores of all parameters with respect to the topology of task T_k

are then summarized as

$$\Omega_k^{(\text{top})} = \left[\|q_r(\mathbf{Z}_k^{[\ell-1]})\| \right]$$

where $\Omega_k^{(\text{top})}$ contains topology-related importance scores of all parameters. In practice, topology scores computed in the intermediate layers of the GNN are used to represent local structural patterns of the blockchain transaction graph.

3) *Final importance score*: The final importance score of parameters Θ for task T_k is computed by combining the loss-based and topology-based components:

$$\Omega_k = \lambda_l \Omega_k^{(\text{err})} + \lambda_t \Omega_k^{(\text{top})} \quad (10)$$

where λ_l and λ_t are hyperparameters that balance the contributions of loss-preserving and topology-preserving terms. Parameters with low importance scores can be updated more freely during new task learning (thus adapting to new forms of blockchain fraud), while stabilizing highly important parameters reduces the risk of catastrophic forgetting of previously observed anomaly patterns.

C. Continual Learning on GNNs

After each anomaly detection task is learned, the proposed TWP module is used to evaluate the importance of all parameters. When learning a new task T_{k+1} , the goal is not only to improve detection performance on the new task, but also to preserve knowledge from previous tasks by maintaining the stability of important parameters. This is achieved by penalizing changes to parameters that were identified as critical for past tasks.

Let Θ_s^* denote the optimal parameters obtained for task T_s and Ω_s the corresponding importance scores. The task-specific loss for training task T_{k+1} is defined as

$$\mathcal{L}_{k+1}^{\text{tot}}(\Theta) = \mathcal{L}_{k+1}^{\text{cur}}(\Theta) + \sum_{s=1}^k \Omega_s \odot (\Theta - \Theta_s^*)^2 \quad (11)$$

where $\odot$ denotes element-wise multiplication, $\mathcal{L}_{k+1}^{\text{cur}}(\Theta)$ is the standard anomaly classification loss on T_{k+1} , and the square is taken element-wise. This strategy ensures that parameters with low importance can be freely updated to adapt to the new fraudulent patterns, while modifications to parameters with high importance incur a larger penalty, thereby preserving knowledge of previously detected blockchain anomalies.

To further enhance model flexibility, the lowest importance scores must also be reduced. While preserving important parameters helps retain knowledge of previous tasks, it risks reducing the model's plasticity and limiting its ability to learn subsequent anomaly types. To preserve sufficient model capacity for future tasks, an ℓ_1 regularization term is applied to minimize the importance scores of parameters computed from the current task. The total loss for training task T_{k+1} is thus given by

$$\tilde{\mathcal{L}}_{k+1}(\Theta) = \mathcal{L}_{k+1}^{\text{tot}}(\Theta) + \beta \|\Omega_{k+1}\|_1 \quad (12)$$

where β is a hyperparameter that controls the capacity reserved for future anomaly detection tasks, and Ω_{k+1} represents the importance scores of all parameters for the current task, computed

using the proposed TWP module. A larger value of β encourages sparser importance patterns, thereby preserving more learning capacity for future blockchain anomalies.

The proposed method can be easily extended to GNNs that do not include an attention mechanism, such as GCN and GIN, making it applicable to arbitrary GNN architectures. A non-parametric attention-like mechanism can be added to construct the topology, formulated as

$$\gamma_{uv}^{[\ell]} = \left(\mathbf{x}_u^{[\ell-1]} \mathbf{U}^{[\ell]} \right)^T \tanh \left(\mathbf{x}_v^{[\ell-1]} \mathbf{U}^{[\ell]} \right) \quad (13)$$

where the score $\gamma_{uv}^{[\ell]}$ depends on the transformed distance or compatibility between node u and its neighbor v in the transaction graph. These scores are then normalized across all neighbors of u to obtain attention-like coefficients. Given the constructed topology, the proposed TWP module can be directly applied to any GNN. It should be noted that while these topology scores are constructed between nodes, they are used solely within the TWP module. The backbone model itself still relies on its original graph structure to aggregate and update node features.

Overall, this continual learning framework enables GNN-based anomaly detection models to adapt to evolving blockchain transaction patterns while mitigating catastrophic forgetting. By explicitly combining loss-based and topology-based importance estimation on transaction graphs, the proposed scheme enhances both adaptability and long-term stability in blockchain security tasks, making it suitable for realistic deployments where new fraudulent behaviors continually emerge over time.

V. EXPERIMENTS

This section evaluates the effectiveness of our scheme on blockchain transaction data. All experiments were conducted in a Python 3.7 environment to ensure stability and compatibility. The model was implemented primarily with PyTorch 1.8.1, running on CUDA 10.1. Supporting libraries included Numpy, Scipy, and Scikit-Learn for numerical computation and machine learning tasks, NetworkX for graph data processing, and Matplotlib for visualization. To enhance graph learning, we also employed the OGB benchmark, Deep Graph Library (DGL), and DGLLife for domain-specific tasks. All experiments were performed on a workstation equipped with an Intel Core i9-10920X CPU @ 3.50 GHz, an NVIDIA RTX 3090 GPU with 24 GB of memory, running Ubuntu 20.04.

For dataset construction, we obtained blockchain transaction data from the CryptoScamDB platform. Among them, nineteen categories of anomalous transaction behaviors with relatively large sample sizes were selected, including labels such as Ponzi schemes and phishing. To facilitate account-level data extraction and organization, we used a Python-based blockchain interaction tool developed by Sokolowska¹. This tool enables the identification of accounts that participate in transactions within a specified block range. By applying it to filter overlapping and duplicate accounts, we obtained the corresponding normal transaction data. Together with the anomalous categories, this

¹ <https://github.com/validitylabs/EthereumDB>

resulted in a total of 20 labeled classes of transaction behaviors, forming a relatively comprehensive and complex dataset for evaluation.

To construct the continual learning tasks, we organized the dataset by anomaly type, where each of the 19 categories of abnormal account behaviors constitutes a distinct anomaly detection task. This design ensures both comprehensiveness and diversity of the continual learning setting. By defining tasks according to different types of fraudulent activities, the experimental framework can evaluate the model's ability to adapt to heterogeneous anomaly patterns while preserving knowledge from previously learned tasks. This construction also enables a more realistic assessment of the proposed framework's long-term stability and adaptability in dynamic blockchain environments.

In the following part, we begin by introducing the evaluation metrics used to assess both anomaly detection accuracy and continual learning stability. Next, we describe the baseline models for comparison and outline the hyperparameter configurations adopted in our experiments. We then report the anomaly detection results, followed by an analysis of catastrophic forgetting to highlight the effectiveness of the proposed approach in retaining past knowledge while adapting to new tasks.

A. Evaluation Metrics

The proposed model addresses two challenges: (i) detecting anomalous blockchain transactions, and (ii) mitigating catastrophic forgetting. Accordingly, we evaluate its performance from two complementary perspectives: anomaly detection and continual learning.

We begin by introducing the notation used throughout this subsection. Let n denote the number of classes. For each class $i \in \{1, \dots, n\}$, let TP_i , FP_i , FN_i , and TN_i represent the numbers of true positives, false positives, false negatives, and true negatives, respectively, measured on the test set.

1) *Anomaly detection*: Since blockchain anomaly detection with GNNs can be formulated as a multi-class classification problem, we adopt standard evaluation metrics commonly used in deep learning: Accuracy, and the Macro/Micro averages of Precision, Recall, and F1-score.

Accuracy measures the proportion of correctly classified samples across the entire test set:

$$\text{Accuracy} = \frac{\sum_{i=1}^n TP_i}{\sum_{i=1}^n (TP_i + FN_i)}. \quad (14)$$

Macro metrics compute per-class scores first and then take the unweighted mean across classes. Specifically, Macro-Pre measures the average precision across classes; Macro-Rec measures the average recall across classes; Macro-F1 combines precision and recall into a single comprehensive score.

$$\text{Macro-Pre} = \frac{1}{n} \sum_{i=1}^n \frac{TP_i}{TP_i + FP_i}, \quad (15)$$

$$\text{Macro-Rec} = \frac{1}{n} \sum_{i=1}^n \frac{TP_i}{TP_i + FN_i}, \quad (16)$$

$$\text{Macro-F1} = \frac{2 \times \text{Macro-Pre} \times \text{Macro-Rec}}{\text{Macro-Pre} + \text{Macro-Rec}}. \quad (17)$$

Micro metrics are sample-weighted averages in which each sample contributes equally. We then compute Micro-Pre, Micro-Recall, and Micro-F1; these reflect the model's performance over individual samples, giving equal weight to every sample.

$$\text{Micro-Pre} = \frac{\sum_{i=1}^n TP_i}{\sum_{i=1}^n TP_i + \sum_{i=1}^n FP_i}, \quad (18)$$

$$\text{Micro-Rec} = \frac{\sum_{i=1}^n TP_i}{\sum_{i=1}^n TP_i + \sum_{i=1}^n FN_i}, \quad (19)$$

$$\text{Micro-F1} = \frac{2 \times \text{Micro-Pre} \times \text{Micro-Rec}}{\text{Micro-Pre} + \text{Micro-Rec}}. \quad (20)$$

2) *Continual learning*: We further evaluate the model with *Average Performance (AP)* and *Average Forgetting (AF)*, which reflect the plasticity–stability trade-off.

Let $M \in \mathbb{R}^{N \times N}$ denote the performance matrix across N sequential tasks, where $M_{i,j}$ represents the performance on task T_j after training up to task T_i .

Average Performance after task T_k is defined as:

$$\text{AP}_k = \frac{1}{k} \sum_{i=1}^k M_{k,i}. \quad (21)$$

AP_k reflects the mean performance over all tasks learned up to step k , capturing the model's overall effectiveness throughout continual training. A higher AP value indicates stronger cumulative learning performance.

Average Forgetting after task T_k is computed as:

$$\text{AF}_k = \frac{1}{k-1} \sum_{i=1}^{k-1} (M_{i,i} - M_{k,i}), \quad 2 \leq k \leq N. \quad (22)$$

AF_k measures the average degradation in performance on previously learned tasks $T_1, \dots, T_{k-1}$ after completing task T_k . For each task T_i , the difference $M_{i,i} - M_{k,i}$ quantifies how much performance has decayed since T_i was first learned. Lower AF values indicate better resistance to catastrophic forgetting.

In summary, Macro/Micro metrics provide a comprehensive evaluation of anomaly detection effectiveness from class- and sample-level perspectives, while AP and AF quantify continual learning stability. Together, they form a unified evaluation framework for assessing both detection accuracy and continual learning robustness.

B. Baseline Models

To validate the anomaly detection performance of our method, we compare it with two representative GNN architectures: the Graph Convolutional Network (GCN) and the Graph Isomorphism Network (GIN). Both are widely used and have shown strong results across various graph-learning tasks:

- 1) *GCN [140]*: Designed to process graph-structured data by aggregating node features from neighbors, GCN effectively leverages both node attributes and structural information.

- 2) *GIN* [141]: A more expressive variant of GNNs, GIN captures complex dependencies and diverse structural patterns, demonstrating strong representational power for heterogeneous graphs.

We also consider Graph Attention Networks (GATs), which improve upon GCNs by applying learnable, feature-dependent attention weights rather than uniform neighbor aggregation. This makes them directly compatible with the *Topology-aware Weight Preservation* (TWP) module. For GCNs, attention coefficients are constructed before applying TWP, while for GINs, node embeddings before pooling are used to compute parameter importance.

To further assess both classification accuracy and resistance to catastrophic forgetting, our method is compared with a bare model (without continual learning) and five state-of-the-art continual learning approaches, all implemented on the same GAT backbone for fairness:

- 1) *Bare model (Bare)*: GNN without continual learning; serves as the lower bound.
- 2) *Elastic Weight Consolidation (EWC)* [122]: Penalizes changes to important parameters to preserve prior knowledge.
- 3) *Memory Aware Synapses (MAS)* [125]: Estimates parameter importance via sensitivity analysis in an online manner.
- 4) *Gradient Episodic Memory (GEM)* [142]: Stores representative samples in memory and constrains gradients to avoid increasing past-task loss.
- 5) *Learning Without Forgetting (LwF)* [123]: Uses knowledge distillation to retain outputs from previous models.
- 6) *Experience Replay GNN (ER-GNN)* [132]: Incorporates replay of representative nodes from prior tasks.
- 7) *Joint Training (Joint)*: Trains on all tasks simultaneously; free from forgetting, serving as the performance upper bound.

C. Hyperparameter Settings

For the algorithms introduced in this subsection, the primary focus is on the optimal settings of the TWP module and the Graph Attention Network (GAT), as these parameters have a crucial impact on both model performance and generalization. Table III shows the hyperparameters in our model.

In the TWP module, λ_l is set to 10000 to ensure stability of the topological structure. The parameter λ_t is chosen within the range [100, 10000]; after experimental evaluation, 1000 is selected as the optimal value, striking a balance between model complexity and performance. The classification performance under different λ_t values is shown in Fig. 2, where the chosen value consistently outperforms alternatives across multiple metrics.

Similarly, β is tuned between 0.1 and 0.01, with the best results achieved at 0.01. The comparison is presented in Fig. 3. This value is applied consistently across all tasks to ensure stable performance.

GAT, a graph neural network based on the attention mechanism, is adopted to process graph-structured data and learn node representations. Its hyperparameters are carefully tuned to optimize performance on the blockchain transaction dataset.

TABLE III
HYPERPARAMETERS OF OUR MODEL

Algorithm	Parameter	Value
TWP	λ_l	10000
	λ_t	1000
	β	0.01
GAT	Number of layers	1
	Hidden units	64
	Attention heads	8
	Output attention heads	1
	Feature dropout rate	0.6
	Attention dropout rate	0.6
	Negative slope	0.2
	Residual connections	No
	Batch size	200
	Learning rate	0.005
	Optimizer	Adam
	Epochs	200
	Node types	40
Train-validation-test split		60%-20%-20%

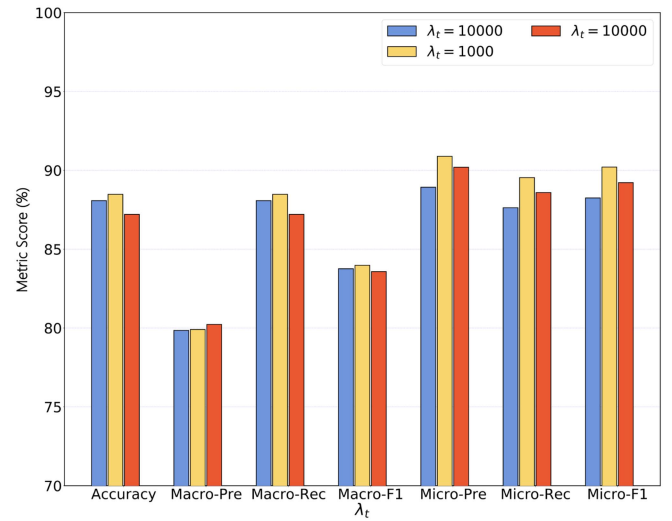


Fig. 2. Effect of different λ_t values on model performance.

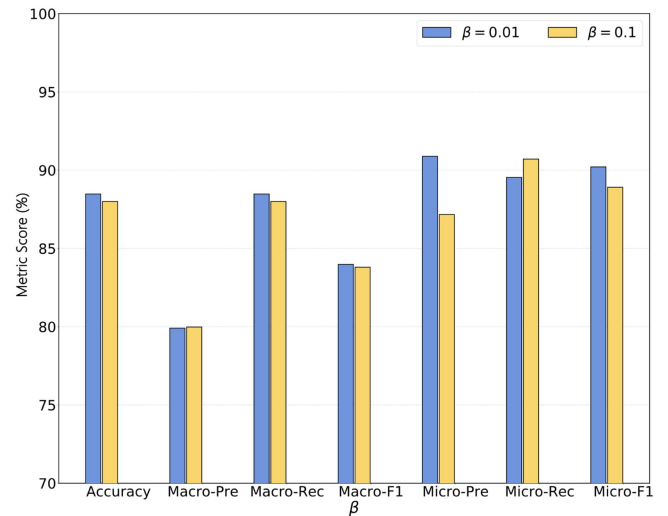


Fig. 3. Effect of different β values on model performance.

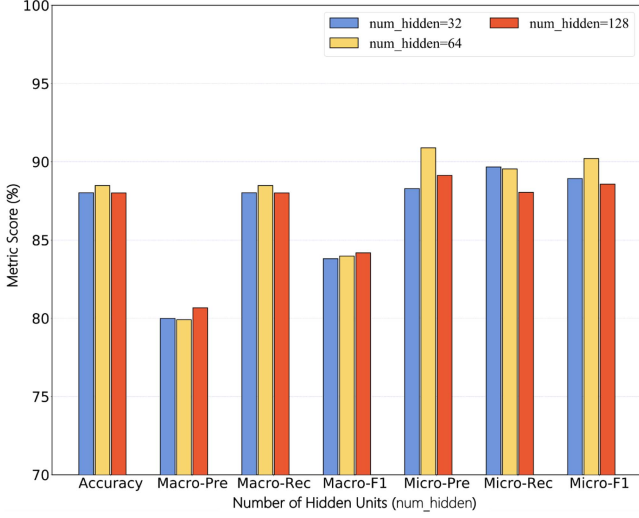


Fig. 4. Impact of hidden unit number on model performance.

First, the number of attention layers is set to one, maintaining simplicity and reducing computational complexity. Second, the number of hidden units in each layer is examined. For high-accuracy classification, this parameter must be carefully chosen: increasing hidden units improves representational capacity but risks overfitting, while too few units may underfit and fail to capture sufficient features. Experiments test 32, 64, and 128 hidden units, as shown in Fig. 4. Results indicate that 64 hidden units generally yield the best performance, outperforming other settings in most metrics. For example, accuracy with 64 hidden units is about 0.5% higher than with the other values. Although 128 units slightly improve Macro-Pre and Macro-F1, the trade-off favors 64, which balances model efficiency and representational power.

In addition, each GAT layer is configured with 8 attention heads, enabling nodes to attend to information from eight different perspectives, thereby enhancing representational richness. The output attention heads are set to 1, meaning only a single aggregated attention vector is used in the final stage.

To prevent overfitting, both feature dropout and attention dropout rates are set to 0.6. The activation function is LeakyReLU with a negative slope of 0.2, which mitigates the vanishing gradient problem and improves learning stability. Residual connections are disabled to simplify the model and avoid unnecessary complexity.

The batch size is fixed at 200, and the learning rate is set to 0.005. The Adam optimizer is selected due to its adaptive learning rate adjustment and fast convergence properties, ensuring stable and efficient training. The number of training epochs is set to 200 to allow sufficient convergence. The dataset is split into 60% training, 20% validation, and 20% test subsets, which is a widely adopted configuration.

Overall, these hyperparameters are carefully tuned to achieve optimal performance and generalization on blockchain transaction datasets, while maintaining a lightweight and interpretable model. They form a solid foundation for subsequent experiments and comparisons.

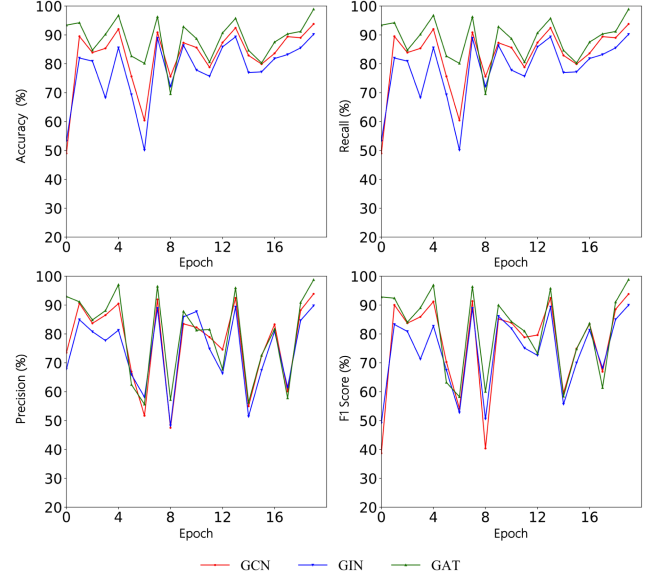


Fig. 5. Comparison of anomaly detection performance across tasks using different GNN backbones. Each subfigure shows the evolution of key metrics (Accuracy, Recall, Precision, and F1 Score) over 20 epochs.

For consistency across baselines, the regularization hyperparameters of EWC and MAS are fixed at 10,000, ensuring a fair comparison under the same experimental conditions. These carefully selected settings guarantee the scientific rigor and reliability of the experiments, while providing valuable references for future research.

D. Anomaly Detection Results

In this subsection, classification tasks are used to evaluate the effectiveness of the proposed model in detecting anomalous blockchain accounts. For comparison, the graph-based deep learning backbone in the proposed model is replaced with other commonly used architectures for anomaly detection, namely GCN and GIN, while keeping all other algorithms and parameters unchanged.

As shown in Fig. 5, we experimentally evaluate the performance of GCN, GIN, and GAT on the blockchain anomalous transaction dataset. Following the evaluation metrics defined before, classification results are obtained for each type of account in every task. The curves in the figure represent accuracy, precision, recall, and F1-score after adding all tasks.

From accuracy and recall, it is clear that, except for a few cases, GAT consistently achieves the highest classification accuracy and recall. The GCN-based model performs better than the GIN-based model overall. In terms of precision, the performance of the proposed model is less stable than for the other two metrics, with results close to those of GCN and GIN on many tasks, but still superior overall. Consequently, for the F1-score, the proposed model with GAT achieves the best performance, with smoother trends than the other two GNN backbones, indicating more stable detection efficiency. Taken together, the multi-task classification results demonstrate the advantage of GAT in modeling complex graph structures and

TABLE IV
EVALUATION METRICS UNDER DIFFERENT GNN BACKBONES

Metric	GCN	GIN	Our Model (GAT)
Accuracy	82.648±0.025	78.048±0.025	88.480±0.017
Macro-Precision	77.376±0.0136	74.640±0.027	79.900±0.017
Macro-Recall	82.648±0.025	78.048±0.025	88.480±0.017
Macro-F1	79.928±0.019	76.310±0.019	83.970±0.017
Micro-F1	83.890±0.039	83.832±0.016	90.126±0.008

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT METHODS

Method	Average Performance AP (%)	Average Forgetting AF (%)
Bare model	53.766±0.037	37.712±0.075
LwF	50.714±0.016	41.746±0.043
GEM	80.924±0.022	9.408±0.032
EWC	80.399±0.039	0.350±0.026
MAS	87.310±0.035	1.182±0.049
ER-GNN	87.236±0.036	3.020±0.038
Our model	88.476±0.014	0.460±0.032
Joint training	89.318±0.056	0.114±0.083

achieving optimal classification across categories. In summary, the GAT-based model outperforms traditional GCN-based approaches for blockchain anomaly detection, and the F1 results further validate the effectiveness and stability of our framework.

Additionally, Table IV presents experimental results using GCN, GIN, and GAT, with Micro-F1 and Macro-F1 adopted as evaluation metrics. The reported values are mean scores, with standard deviations shown in parentheses. From the perspective of Micro-F1, the three GNN backbones achieve 83.89%, 83.832%, and 90.126%, respectively. These results indicate that GAT achieves the highest classification accuracy, as it is able to capture higher-order structural information and richer meta-paths. Similarly, from the perspective of Macro-F1, the models achieve 79.928%, 76.31%, and 83.97%, respectively, again showing that GAT performs best. The trends are consistent with those observed in Micro-F1.

In summary, consistent with the findings in Fig. 5, GAT demonstrates superior performance on classification tasks involving complex graph structures, thereby validating the effectiveness of the proposed model. Moreover, the relatively small standard deviations indicate stable performance across tasks.

E. Catastrophic Forgetting

In anomalous transaction detection on blockchains, quantitative comparison is essential for evaluating the strengths and weaknesses of different continual learning strategies. The results of several baseline methods and our proposed approach are summarized in Table V, which reports two key evaluation metrics: AP and AF. AP measures the average classification accuracy across all learned tasks, reflecting the model's overall detection capability and adaptability to new anomalous behaviors. In contrast, AF quantifies the average degradation in performance on previously learned tasks after the model learns new ones, thereby indicating the degree of *catastrophic forgetting*. In general, higher AP and lower AF jointly represent stronger continual

learning ability, striking a balance between plasticity (adaptation to new tasks) and stability (retention of prior knowledge).

Table V presents the comparative results across seven continual learning methods and a joint training upper bound. Each value in the table represents the mean and standard deviation over multiple runs. The bare model achieves moderate AP but exhibits the highest AF, confirming that standard GNNs suffer severe catastrophic forgetting when trained sequentially. Among continual learning baselines, methods such as GEM and MAS effectively mitigate forgetting and substantially improve AP. However, EWC, despite achieving the lowest AF (0.35%), sacrifices plasticity, resulting in nearly an 8% drop in AP compared with our method. This trade-off highlights that minimizing forgetting alone does not ensure high overall performance, and that a successful continual learner must simultaneously preserve past knowledge and remain responsive to new data.

Our proposed model achieves the best balance, ranking first in AP (88.48%) and maintaining one of the lowest AF values (0.46%). These results demonstrate that the TWP mechanism enables the model to retain learned representations while continually integrating novel patterns, thereby addressing the stability-plasticity dilemma more effectively than other baselines.

The joint training strategy, which learns all tasks simultaneously using full access to past data, serves as an upper-bound reference. Although it achieves slightly higher AP (89.32%) and minimal AF (0.11%), such an approach is infeasible in practical continual learning scenarios where data from previous tasks are not accessible. The close performance of our model to this upper bound confirms its scalability and robustness for real-world blockchain anomaly detection under realistic sequential learning conditions.

Fig. 6 further illustrates the performance trend of the first task as the number of learned tasks increases. As expected, performance on the first task gradually declines with the introduction of new tasks, which is an intuitive manifestation of catastrophic forgetting. Some baselines (e.g., ER-GNN) occasionally outperform our model on the first task, but their results fluctuate significantly, reflecting unstable detection efficiency. By contrast, the proposed model exhibits more stable behavior, with a much slower decline in performance than competing methods.

Fig. 7 presents the evolution of average performance across all tasks. Similar to Fig. 6, performance generally declines as new tasks are introduced. Nevertheless, our method consistently achieves the lowest AF among all approaches, with performance levels approaching the theoretical upper bound. This demonstrates its ability to effectively preserve knowledge from earlier tasks while maintaining strong adaptability to new ones.

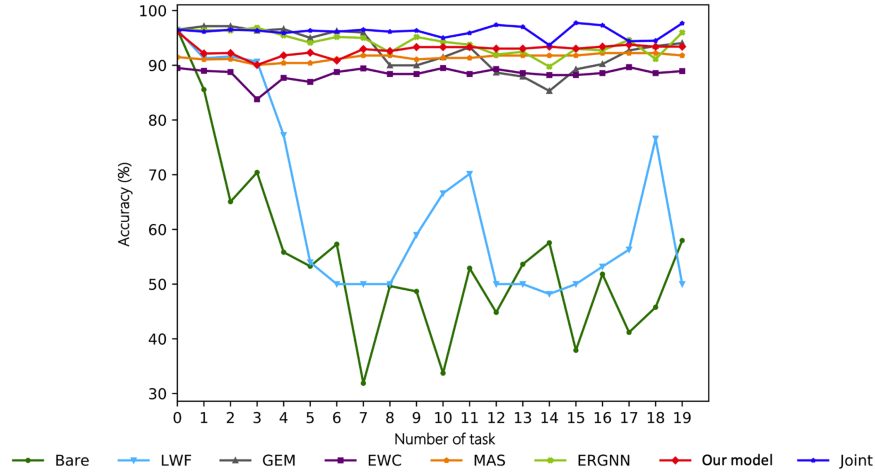


Fig. 6. Average performance on the first task.

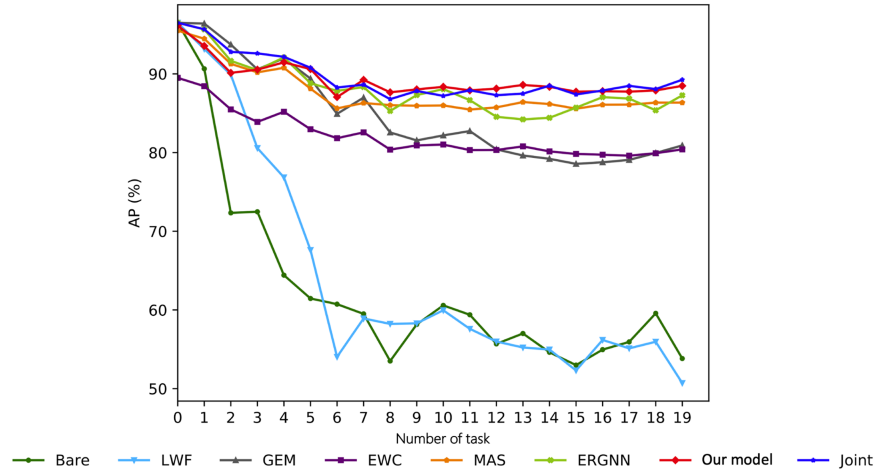


Fig. 7. Average performance across all tasks.

Among the baselines, LwF is a commonly used approach that seeks to mitigate forgetting by expanding network layers with new nodes for each task. However, as shown in Fig. 7, its performance is often close to the lower bound for multi-class node classification, indicating limited adaptability of LwF to GNNs. Similarly, GEM employs memory replay by storing subsets of past data, but its performance is unstable and occasionally drops sharply, likely due to inefficient or rigid replay strategies.

In contrast, EWC and MAS, both of which constrain updates to important parameters, deliver stronger results, confirming the effectiveness of protecting critical parameters as a means of reducing forgetting. Building on this idea, we further incorporate topology-aware parameter importance into these methods, which enhances their performance and strengthens their ability to address catastrophic forgetting in GNNs.

In summary, our scheme achieves superior performance across multiple metrics on the blockchain anomaly detection dataset. Detailed analysis of experimental results confirms that the method offers significant advantages in mitigating forgetting, preserving knowledge from previous tasks, and maintaining strong performance when learning new tasks.

VI. DISCUSSION

The experimental results presented in Section V demonstrate the effectiveness of the proposed graph-based continual learning framework for blockchain anomaly detection. This section provides a deeper interpretation of these results, discusses their practical implications, acknowledges the limitations of our current work, and outlines potential directions for future research.

A. Interpretation of Results

The superior performance of our model, particularly when using GAT as the backbone network, can be attributed to the synergistic effect of its core components. The attention mechanism inherent in GAT allows the model to dynamically weigh the importance of neighboring nodes and transactions, which is crucial for identifying subtle and complex fraudulent patterns that often involve coordinated activities across multiple accounts. This capability is especially valuable in blockchain ecosystems, where the relational structure between entities carries significant signals for anomaly detection.

More importantly, the introduced Topology-aware Weight Preservation (TWP) module successfully addresses the core challenge of catastrophic forgetting in continual learning. By estimating parameter importance based on both task-specific loss and local topological structures, the TWP module provides a more nuanced and effective regularization strategy than methods that consider only loss gradients (e.g., EWC) or generic parameter sensitivity (e.g., MAS). The results in Table V and Figs. 6–7 confirm that our approach achieves an excellent balance between stability (low AF) and plasticity (high AP), outperforming other state-of-the-art continual learning methods. This indicates that explicitly modeling the graph topology is not only beneficial for the primary detection task but also critical for preserving knowledge in a sequential learning setting.

B. Practical Implications

The proposed framework provides a scalable and deployable solution for real-world blockchain security monitoring. In operational environments, new types of attacks and fraudulent schemes continually emerge, while transaction volumes grow rapidly. The capability to incrementally learn novel anomalous patterns, without retraining from scratch on the entire historical dataset, offers substantial advantages in both computational efficiency and resource utilization.

Experimental evaluations on 19 distinct categories of blockchain risk behaviors demonstrate that the proposed approach can not only detect well-known threats such as phishing and Ponzi schemes, but also adaptively learn the characteristics of newly emerging risks. This continual learning ability enables the framework to evolve alongside adversarial tactics, thereby enhancing its practicality for long-term, adaptive blockchain anomaly detection and contributing to sustained ecosystem protection.

The proposed method primarily targets the identification of abnormal accounts in account-based blockchains such as Ethereum, where nodes represent accounts and edges correspond to transactions. To extend the approach to UTXO-based systems (e.g., Bitcoin), an additional address-aggregation process is required to map multiple addresses controlled by the same entity into a single node. This can be accomplished by constructing an entity graph that groups related addresses likely belonging to the same user, as described in prior studies [54], [143]. After entity aggregation, the resulting structure becomes conceptually equivalent to the Ethereum account graph, allowing the same continual learning framework to be applied for anomaly detection. When transferring to UTXO-based systems, Ethereum-specific features (e.g., smart contract attributes) should be excluded from training to ensure consistency and comparability across blockchain types.

From a deployment perspective, the framework exhibits three key advantages. First, its modular architecture allows integration with various GNN backbones (e.g., GCN, GIN, GAT), enabling flexible adaptation to diverse blockchain infrastructures with minimal architectural changes. Second, the incremental update process significantly reduces retraining costs, making it suitable for security systems that require frequent updates on continuously streaming blockchain data. Third, the design is

compatible with distributed or federated training settings, which could further enhance scalability when monitoring multiple blockchain networks simultaneously.

However, large-scale deployment also introduces challenges. Maintaining real-time performance under high transaction throughput may require distributed graph processing and optimized GPU scheduling. Moreover, as the number of incremental tasks grows, the accumulated parameter regularization terms could increase memory consumption. These aspects suggest that practical deployments may benefit from adaptive pruning or periodic model compression strategies.

C. Limitations and Future Directions

While the proposed framework shows promising results, our study has several limitations that also chart a course for future research.

- *Task-Incremental Setting:* Our current framework operates in a task-incremental learning setting, where task boundaries are clearly defined and known during training. However, in a real-world deployment, anomalies may emerge in a more blurred and task-agnostic stream. To overcome this limitation, future work will focus on developing a task-free or class-incremental adaptation of our topology-aware preservation strategy, enabling effective learning from a continuous, non-stationary stream of transaction data without explicit task boundaries.
- *Dependency on Label Quality:* The performance of the model is inherently tied to the quality and timeliness of the labels used for each task. Label scarcity and potential labeling noise in blockchain data remain persistent challenges. To address this, we plan to explore integrating self-supervised pre-training and semi-supervised learning techniques into our continual learning paradigm. This would mitigate the dependency on large amounts of meticulously labeled data for each new task, enhancing the framework's robustness and practicality.
- *Computational Overhead and Scalability:* Although the proposed approach avoids full retraining, calculating topology-aware importance scores still introduces non-negligible computational overhead. For high-throughput or multi-chain monitoring scenarios, this may affect real-time responsiveness. Future work will investigate efficient approximations for the TWP module, such as importance sampling, sparse parameter updates, or low-rank approximations, to reduce memory and time costs. Additionally, integrating the method into parallelized graph processing frameworks could further enhance scalability for industrial deployment.
- *Potential Failure Cases:* The framework assumes that the graph structure sufficiently captures relationships relevant to fraud detection. In scenarios where malicious behavior is intentionally obfuscated or dispersed across unrelated transactions, graph-based correlations may weaken, leading to reduced detection sensitivity. Exploring hybrid architectures that combine graph-based learning with temporal or semantic anomaly modeling represents an important future direction.

In conclusion, the discussion highlights that our proposed method is not only effective but also lays a foundation for building more adaptive, efficient, and robust security systems for the evolving blockchain landscape. The identified limitations provide a clear roadmap for future research to enhance the framework's practicality and performance.

VII. CONCLUSION

This paper presented a graph-based continual learning framework for anomalous account detection in blockchain systems. The proposed approach achieves high detection accuracy while effectively mitigating catastrophic forgetting, a long-standing challenge in dynamic environments. At its core, the Topology-Aware Weight Preservation module captures structural dependencies within transaction graphs and evaluates parameter importance from both task-specific and topological perspectives. By constraining updates to critical parameters, the model preserves past knowledge while adapting to new transaction patterns.

Beyond its empirical success, the framework demonstrates strong potential for real-world blockchain security deployment. Its modular architecture supports integration with various GNN backbones, and its incremental update mechanism substantially reduces retraining costs, enabling continuous monitoring over large-scale blockchain data streams. Nonetheless, practical scalability introduces additional challenges, including the need for distributed processing, efficient memory management, and low-latency inference under high transaction throughput. Addressing these challenges is key to bridging the gap between research prototypes and production-ready systems.

Future work will explore task-free or class-incremental continual learning to handle continuously evolving transaction streams without explicit task boundaries, as well as self-supervised and semi-supervised extensions to alleviate reliance on labeled data. Moreover, hybrid architectures that combine graph-based structural reasoning with temporal and semantic representations may further enhance robustness against obfuscated or cross-chain fraudulent activities.

In summary, this study establishes a foundation for continual, topology-aware graph learning in blockchain anomaly detection, advancing toward scalable, adaptive, and intelligent security systems capable of protecting decentralized ecosystems in the long term.

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